

Design Project — Exam Revision Pack

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Course map: L1 Introduction to Projects · L2 The Project Cycle & Models · L3 Project Identification · L4 PM Disciplines · L5 Project Writing (Proposal & Logframe) · L6 Project Evaluation.

This pack: Part A = core theory · Part B = full model answers to the four tutorial exercises (which are scenario-based — the exam format).

Part A — Core Theory

1. Project fundamentals

Project = a **temporary endeavour undertaken to create a unique product, service or result** (PMI). Has a clear start & end and produces deliverables within constraints of scope, time, cost & quality.

Key characteristics: Uniqueness · Temporality · Objective (measurable goal) · Resources · Constraints · Uncertainty & Risk.

Hierarchy: Portfolio > Program > Project. **Triple Constraint (Iron Triangle): Scope · Time · Cost** (Quality often the 4th) — interdependent, changing one affects the others.

Engineering Design Process (iterative): 1 Problem Definition/Needs → 2 Conceptual Design → 3 Feasibility Analysis → 4 Embodiment (Preliminary) Design → 5 Detailed Design (CAD, BOM) → 6 Prototyping & Testing (V&V) → 7 Communication/Handover.

2. The Project Cycle & models

Standard phases: Identification → Preparation/Planning → Appraisal & Financing → Implementation → Monitoring & Evaluation → Closure.

EU PCM (6 phases)	PMBOK / PMI (4)	Development Bank (WB / AfDB)
Programming · Identification · Formulation · Financing · Implementation · Evaluation	Initiation · Planning · Execution & Monitoring · Closure	Country Strategy · Identification · Preparation & Appraisal · Negotiations & Board Approval · Implementation & Supervision · Post-Completion Evaluation

3. Project Identification — the Problem Tree

The **Problem Tree** identifies the **core problem**, its **causes** (roots) and **effects** (branches). It is converted into an **Objectives Tree** (means-ends): core problem → main objective; causes → means/activities; effects → ends/results. Output of this phase = the **Project Concept Note (PCN)**.

4. The six PM disciplines (Dr MougnoI's core set)

Scope · Time · Resource · Risk · Justification · Stakeholder Management. (Risk responses = **Avoid, Transfer, Mitigate, Accept**; Scope creep = uncontrolled scope expansion.)

5. Project Writing — Proposal & Logframe

Logframe (4×4 matrix) rows = **Goal → Purpose → Outputs → Activities**; columns = **Narrative summary · Indicators · Means of Verification · Assumptions**.

6. Project Evaluation — OECD-DAC six criteria

Criterion	Key question
Relevance	Is the project doing the right things (suited to needs/priorities)?
Coherence	Does it fit with / complement other interventions?
Effectiveness	Has it achieved its objectives?
Efficiency	Has it used resources/inputs well (cost-effective)?
Impact	What broader long-term difference has it made?
Sustainability	Will the benefits continue after funding ends?

Part B — Tutorial Exercises: Model Answers

Exercise 1 — New Cybersecurity BTech programme at IUGET Douala

(a) Define "project" + FIVE characteristics with evidence from the scenario.

Project = a temporary endeavour undertaken to create a unique product, service or result (PMI).

Characteristic	Evidence from the scenario
Uniqueness	The Cybersecurity BTech "has never been undertaken before" at IUGET — a one-off outcome.
Temporality	Launching a programme has a defined start and a target launch date (an end).
Objective	A specific measurable goal: establish and launch the new programme.
Resources	Needs lecturers, funding, labs/equipment and a curriculum.
Constraints	Bound by budget (cost), the academic calendar (time) and curriculum/accreditation (scope).

(A 6th valid point: Uncertainty/Risk — being a first-ever initiative it carries high risk.)

(b) Project vs Routine Operations — THREE differences (table).

Project	Routine Operations
Temporary — has an end date	Ongoing / continuous
Produces a unique output	Produces repetitive output
Higher risk & uncertainty; cross-functional team	Lower risk, predictable; permanent departmental team

(c) Triple Constraint applied.

The **Iron Triangle** balances **Scope** (curriculum, labs, accreditation requirements), **Time** (ready to admit students by a set semester) and **Cost** (the approved budget). They are interdependent: e.g. adding more specialised lab equipment (scope↑) raises cost and may delay the launch (time↑); forcing an earlier launch (time↓) may require more money or a reduced curriculum (scope↓). Quality (accreditation standard) sits at the centre and must be protected.

Exercise 2 — AfDB rural electrification, Adamawa Region

(a) FOUR standard phases of the project cycle (purpose · activities · outputs).

Phase	Purpose	Key activities	Output
Identification	Decide if the project is worth pursuing	Situation & problem analysis, feasibility concept	Project Concept Note
Preparation / Formulation	Plan all activities, resources, timelines, budget in detail	Detailed design, Logframe, stakeholder consultation	Full project proposal/document
Appraisal & Financing	Assess viability & secure funds	Technical/financial/economic/social/environmental appraisal, approval	Financing agreement / approval
Implementation (& M&E)	Deliver the project per plan	Mobilise resources, build grid, monitor progress	Electrified communities; progress reports

(b) AfDB/World Bank cycle vs EU PCM — THREE similarities, TWO differences.

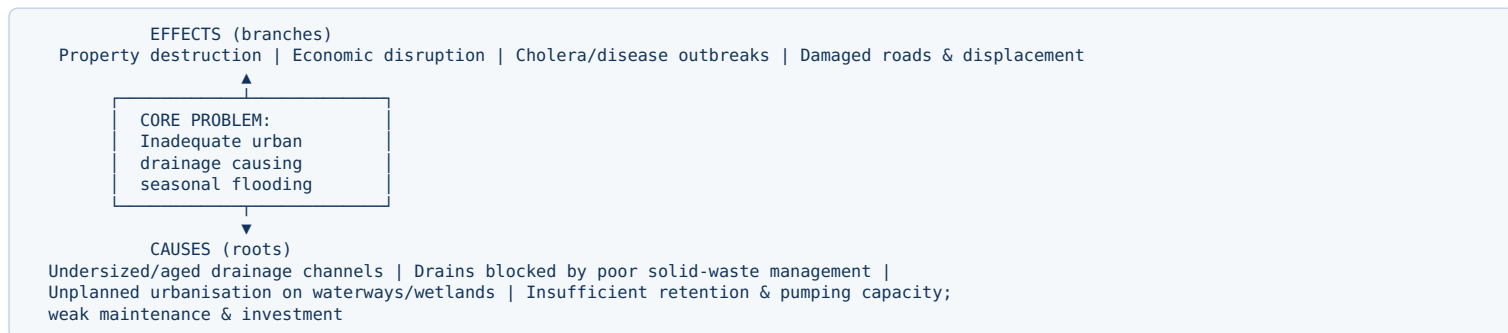
Similarities: (1) both are cyclical/phased frameworks from identification to evaluation; (2) both use appraisal and the **Logical Framework** in design; (3) both stress monitoring, accountability and learning. **Differences:** (1) EU PCM has **six named phases** starting with **Programming** (strategic orientation), whereas the AfDB/WB cycle starts with a **Country Strategy** and is tailored to large infrastructure; (2) the AfDB/WB cycle includes an explicit **Negotiations & Board Approval** step that the EU PCM does not.

(c) Why project cycle models matter for development organisations.

They give **systematic planning & scheduling**, ensure **consistent decision-making** at each phase (go/no-go gates), enforce **accountability & transparency** in the use of donor funds, and enable **learning** from completed projects to improve future ones.

Exercise 3 — Douala (CUD) drainage & flooding project

(a) Problem Tree.



- **Core problem:** Inadequate/failing drainage infrastructure causing seasonal flooding across 14 neighbourhoods.
- **Four root causes:** undersized/ageing channels · drains blocked by uncollected solid waste · unplanned building on natural waterways · lack of retention/pumping capacity and maintenance.
- **Four effects:** destruction of property · disruption of economic activity · cholera outbreaks · flood-damaged roads / population displacement.

(b) SIX PM disciplines applied to the drainage project.

Discipline	Definition	Application to the Douala project
Scope	Ensures all & only the required work is done	Define the 85 km channels, 12 basins, 6 pumping stations, 38 km roads; control scope creep
Time	Plan & control the schedule	Sequence the 5-year works with a Gantt chart around the rainy season
Resource	Plan people, materials, equipment & budget	Allocate the XAF 68 bn, machinery, labour and materials efficiently
Risk	Identify, analyse & respond to risks	Mitigate flood-season delays, land-acquisition and environmental risks
Justification	Maintain the business case/benefit	Justify the cost by reduced flooding, cholera and economic loss
Stakeholder	Engage all affected parties	Manage CUD, World Bank, the 14 neighbourhoods, contractors & government

(c) Risk Register (five risks).

No.	Risk Description	Type	Prob.	Potential Impact	Response Strategy
1	Delay in acquiring the 4.2 km ² of land	Legal/social	High	Schedule slippage, cost overrun	Mitigate — start acquisition & resettlement early
2	Flooding/heavy rain halts construction	Environmental	High	Work stoppage, damaged works	Mitigate — schedule critical works in dry season
3	Cost overrun / funding shortfall	Financial	Medium	Incomplete scope	Transfer/Mitigate — contingency budget, fixed-price contracts
4	Contractor underperformance	Technical/management	Medium	Poor quality, delays	Mitigate — prequalify contractors, supervise, penalties
5	Community resistance / disputes	Social	Medium	Protests, stoppages	Mitigate — stakeholder engagement & compensation

Exercise 4 — EnergyAccess Cameroon solar mini-grid proposal

(a) EIGHT proposal sections (purpose + key element).

Section	Purpose & key element
1. Title Page	Identifies the project — title, organisation, date, location, project manager
2. Executive Summary	1-2 page overview — problem, objectives, results, budget, duration
3. Background / Context	Justifies the need — data/evidence on lack of electricity in Mbam-et-Inoubou
4. Problem Statement	Clear statement of the problem — linked to the problem analysis
5. Objectives	Goal + specific objectives — electrify 3,000 households & 20 health centres
6. Activities / Work Plan	List of activities & timeline — often a Gantt chart
7. Budget	Line-item costs — the XAF 2.5 bn incl. contingency & overhead
8. Logframe + M&E	Project logic & how results are tracked — indicators & reporting

(Also valid: Outputs, Risk Analysis, Organisational Capacity, Annexes.)

(b) Logframe matrix (solar mini-grid).

Level	Narrative	Indicator	Means of Verification	Assumptions
Goal	Improved living standards & economic activity in Mbam-et-Inoubou	% rise in local income/health access	Regional statistics,	Stable governance & security

			surveys	
Purpose	Reliable electricity for 3,000 households & 20 health centres	No. of connections with power ≥ 16 h/day	Meter records, utility reports	Users pay tariffs; grid maintained
Output 1	Solar mini-grid installed & operational	kW capacity installed	Commissioning report	Equipment delivered on time
Output 2	Households & clinics connected	No. of connections completed	Connection register	Community cooperation
Activities	1 Site survey · 2 Procure & install panels/batteries · 3 Lay distribution lines · 4 Connect users & train operators	Inputs: engineers, panels, XAF 2.5 bn	Budget & procurement records	Funds released; skilled labour available

(c) OECD-DAC six criteria — evaluator's question for this project.

- **Relevance:** Does the solar mini-grid match the community's real energy needs and national priorities?
- **Coherence:** Does it fit with other rural-electrification/World Bank interventions in the region?
- **Effectiveness:** Were the 3,000 households and 20 health centres actually electrified?
- **Efficiency:** Was the XAF 2.5 bn used cost-effectively and on time?
- **Impact:** What broader change resulted (health, education, income)?
- **Sustainability:** Will the mini-grid keep running after donor funding ends (tariffs, maintenance)?